IoT-Based Smart Medication Adherence System

Project report submitted in partial fulfilment of the requirements for the course- Embedded System and Internet of Things (23IC002) of

Bachelor of Engineering

in

Computer Science and Engineering

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May, 2025

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNJAB

(Accredited by NAAC with Grade 'A+')

DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

CERTIFICATE

This is to certify that the project titled "**IoT-Based Smart Medication Adherence System**" submitted to the **Chitkara University Institute of Engineering and Technology (CUIET)** by **Tejinder Kaur 2320992230, Vidhi Joshi, 2320992242, Vivek, 2320992245,Aaina, 2320992595, Shivek S Mittal, 2320992527** is a bonafide record of the work done by the students towards partial fulfilment of requirements for the course- Embedded System and Internet of Things (23IC002) of **Bachelor of Engineering in Computer Science and Engineering.**

	Supervisor/Project Guide
Signature	
Name	
Designation	
Department	**DICE**

Place: Chitkara University, Rajpura, Punjab

Date:

Candidates' Declaration

We, **Tejinder Kaur 2320992230, Vidhi Joshi, 2320992242, Vivek, 2320992245, Aaina, 2320992595, Shivek S Mittal, 2320992527**, of **Group- 25-B**, B.E. -2023 batch of Chitkara University, Punjab hereby declare that the Embedded System and Internet of Things (ES&IoT) project entitled **"IoT-Based Smart Medication Adherence System"** is an original work and data provided in the project report is authentic and to the best of our knowledge. This project has not been submitted by us to any other institute for the award of any other course.

Contribution Details:

Sr. No.	Student Name	Roll No.	Contact Number	Contribution in Project	Signature
1.	Tejinder kaur	2310992230	9877934512	Led the team, coordinated tasks, and managed final integration and documentation.	
2.	Vidhi Joshi	2310992242	9306806434	Handled circuit design, sensor wiring, and hardware assembly.	
3.	Vivek Gusian	2310992245	7888695794	Tested hardware and software, ensuring reliable system performance.	
4.	Shivek S Mittal	2310992527	9466483534	Developed and debugged Arduino code and IoT functionalities using Blynk.	
5	Aaina Kaushik	2310992595	7986146285	Researched related work and prepared the complete project documentation.	

ACKNOWLEDGEMENT

Completion of this project would not have been possible without the help of many people, to whom we are thankful. First, we would like to convey our sincere thanks to **Dr. Rajneesh Talwar**, **Professor and Dean**, DICE-CUIET, Chitkara University, Punjab, for his constant support, motivation and encouragement towards projects and proving healthy environment for working on real time problems.

We express our sincere gratitude to **Dr. Rishu Chhabra, Cluster Dean,** CSE-IInd year, for her constant support and cooperation.

We would like to express our sincere gratitude to our supervisor**, Dr. Amandeep Kaur (Assistant Professor) ,** Her constant motivation, guidance and support helped us a great deal to achieve this feat.

We would like to thank **Dr. Manvinder Sharma, Associate Professor,** Course Coordinator-ES&IOT, DICE, for guiding and inspiring us in many ways. We are also thankful to other faculty members and staff of the Department of Interdisciplinary Courses in Engineering and Computer Science and Engineering department for their support.

We also thank our staff **Ms. Shubhi** and **Mr. Jaswinder Singh,** who helped us in lab experiments throughout the semester.

We wish a deep sense of gratitude and heartfelt thanks to the management for providing excellent lab facilities and tools.

PROBLEM STATEMENT

Medication non-adherence is a significant and persistent issue in healthcare, often leading to poor treatment outcomes, increased healthcare costs, and preventable hospitalizations. This challenge is especially critical among elderly individuals, patients with chronic illnesses, and those managing complex medication regimens. Many of these patients struggle with remembering doses, managing multiple prescriptions, or understanding their medication schedules.

One of the core reasons for non-adherence is the absence of real-time monitoring and personalized reminders. Traditional approaches such as pill organizers or manual check-ins are insufficient, as they do not provide immediate feedback or continuous support. As a result, healthcare providers often remain unaware of missed doses until adverse effects become evident.

There is a pressing need for a technology-driven solution that can support patients in adhering to their prescribed treatments. An Internet of Things (IoT)-based system offers a promising approach, enabling smart monitoring, automated reminders, and timely alerts for both patients and caregivers. By integrating IoT into medication management, it is possible to significantly reduce non-adherence, improve patient outcomes, and minimize unnecessary healthcare interventions. Such a solution can also enhance patient independence and offer peace of mind to families and medical professionals.

ABSTRACT

Abstract:

Medication non-adherence remains a critical challenge in healthcare, particularly among elderly individuals, patients with chronic conditions, and those managing complex medication regimens. This issue leads to poor treatment outcomes, increased healthcare costs, and preventable hospitalizations. Traditional approaches, such as pill organizers and manual check-ins, are inadequate in providing real-time monitoring and personalized reminders. As a result, missed doses often go unnoticed by healthcare providers until adverse effects occur. To address this, an Internet of Things (IoT)-based solution offers a promising approach by integrating smart monitoring, automated reminders, and timely alerts for both patients and caregivers. This technology-driven system can significantly reduce medication non-adherence, enhance patient outcomes, and minimize unnecessary healthcare interventions, while also promoting patient independence and providing peace of mind to families and medical professionals.

This low-cost, scalable solution bridges the gap between simple pillboxes and expensive health monitoring systems, promoting better health adherence using accessible communication technologies.

Application Area(s) of the Project: The IoT-based medication management system is highly applicable in the following areas:

1. **Healthcare Management:**
    - Assists in improving medication adherence for patients, especially those with chronic illnesses, the elderly, or individuals managing multiple prescriptions.
    - Helps healthcare providers monitor patient medication adherence remotely, enabling timely intervention if necessary.
2. **Elderly Care:**
    - Supports elderly individuals who may have difficulty remembering their medication schedules, enhancing their independence while reducing the risk of missed doses.

3. **Chronic Disease Management:**
    - Assists patients with chronic conditions (such as diabetes, hypertension, etc.) in maintaining consistent medication routines, improving long-term health outcomes.
4. **Home Healthcare:**
    - Provides caregivers and family members with real-time notifications and alerts, ensuring that patients adhere to prescribed treatments without requiring constant oversight.
5. **Telemedicine and Remote Monitoring:**
    - Integrates with telemedicine platforms, allowing healthcare providers to remotely monitor medication adherence and communicate with patients without in-person visits.
6. **Pharmaceutical Industry:**
    - Offers potential for pharmaceutical companies to integrate adherence-tracking features into their products, ensuring proper usage and maximizing the effectiveness of treatments.

Technology Stack:

IoT Devices:

- **ESP32 for device connectivity, handling the interaction between the medication box and the cloud.**
- **IR Sensor to monitor whether the medication is taken correctly.**

Notifications:

- **Blynk for sending notifications and alerts to patients, caregivers, and healthcare providers regarding missed or taken medications.**

Sustainable Development Goals (SDGs) Covered: Write SDG goals related to your proposed project

eg. SDG 9- Industry, Innovation and Infrastructure

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

1.1 Project Overview

The aim of this project is to design and implement an IoT-based **Smart Medication System** that enhances medication adherence by providing real-time monitoring, automated visual reminders, and mobile notifications. The system is especially beneficial for elderly individuals, patients with chronic illnesses, and those managing complex medication schedules. It leverages the **ESP32 microcontroller**, **IR sensor** and the **Blynk app** to create a responsive, connected, and user-friendly solution.

By combining sensor inputs, wireless communication, and intuitive alerts, the system ensures that users are reminded to take their medicine and that caretakers can monitor compliance remotely. The project highlights how IoT can be applied to healthcare to improve treatment outcomes and support independent living.

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1.2 Problem Statement

The following problems in healthcare were identified, prompting the development of this project:

- Patients often forget or delay their medication, leading to poor health outcomes.
- Manual reminder systems are ineffective, especially for the elderly or cognitively impaired.
- There is a lack of real-time monitoring or alert systems to notify caregivers of missed doses.
- Medication non-adherence contributes to increased healthcare costs and hospitalizations.

1.3 Proposed Solution

To tackle these issues, an affordable and user-friendly IoT-based medication monitoring system is proposed. Key features include:

- **Real-Time Monitoring:** An IR sensor detects user interaction to confirm dose intake.
- **Mobile Alerts:** The Blynk app sends notifications to users and caregivers about medication schedules and missed doses.
- **Remote Access:** Caregivers can monitor and adjust reminders from any location.

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1.4 Objectives of the Project

- To ensure timely medication through automated alerts.
- To provide real-time monitoring of dose adherence using sensors.
- To enable remote scheduling and notifications via the Blynk app.
- To assist elderly or chronically ill patients with an intuitive and reliable system.
- To offer a scalable, cost-effective healthcare solution for home-based settings.

1.5 Significance of the Project

This project addresses the growing need for effective medication management solutions by merging technology with everyday healthcare challenges. It empowers patients and caregivers alike by reducing human error, increasing adherence, and supporting independent living. Moreover, by making the system affordable and IoT-enabled, it paves the way for widespread adoption across homes and care facilities.

1.6 General Working

The system uses an **ESP32 microcontroller** as its core, connected to an **IR sensor** for detecting user interaction, a buzzer and a speaker for beep that tells you that It's time for medicine and a button that's would record medicine taken or not if and only if you pressed the button. The **Blynk app** allows for mobile-based control and scheduling. When the time for medication arrives, the system displays a reminder on the LCD and checks for user response via the IR sensor. If no action is detected within a preset time and button is not pressed, a notification is sent to the caregiver or user via the Blynk app, ensuring timely intervention and adherence.

CHAPTER 2

PROJECT WORK

2.1 Problem Identified

In today's world, managing health efficiently is becoming increasingly important, especially for elderly individuals and patients with chronic conditions. Despite this, medication non-adherence remains a significant problem. Key challenges include:

- **Missed or incorrect doses** due to forgetfulness or confusion, especially among the elderly.
- **No real-time monitoring**, making it difficult for caregivers to track whether medication was taken.
- **Lack of notifications**, meaning patients do not receive timely reminders.
- **Manual systems** (e.g., alarms or written schedules) are not dynamic or interactive.

These issues often lead to poor treatment outcomes, increased healthcare costs, and avoidable hospitalizations. Therefore, there was a need for an affordable, smart solution to help patients take their medications on time with minimal oversight.

2.2 Project Overview

To address this issue, I developed an **IoT-Based Smart Medication System** using the **ESP32 microcontroller**, **IR sensor**, **Buzzer, Button**, and the **Blynk app**. The system provides **visual reminders**, **monitors user response in real time**, and **sends notifications** to caregivers or users via a mobile app.

Users receive reminders beeping through Buzzer at scheduled times set through the app. When a hand or object is detected near the pillbox (via the IR sensor) and button is pressed, the system records the response as dose taken. If no response is detected, an alert is sent via Blynk, ensuring follow-up action.

2.3 How the Problem Was Solved

The solution was built using the following approach:

1. **Hardware Setup:**
   - ESP32 microcontroller was used for control and Wi-Fi connectivity.
   - An IR sensor detects interaction with the medication box.
   - A buzzer/Speaker that beeps at scheduled time.
   - A button that's must for recording of you medicine taken or not
2. **Software Development:**
   - The Blynk app was configured to send notifications and manage schedules.
   - The system logic was programmed using Arduino IDE and uploaded to the ESP32.
3. **Testing and Deployment:**
   - The system was tested for detection accuracy, notification timing, and user response logging.
   - Alerts and sensor readings were verified under different scenarios to ensure reliability.

2.3 Uniqueness of the Project

This project stands out for its simplicity, low cost, and focus on healthcare. Unlike commercial pill dispensers, it is **modular**, **easily customizable**, and uses **open-source** components. Most importantly, it combines **reminders**, **sensor-based interaction tracking**, and **remote alerts**—features that are rarely all present in a single low-cost solution.

2.4 Challenges Faced

Several issues had to be addressed during development:

- **Wi-Fi Reliability:** Unstable connections required retries and fallback logic.
- **Sensor Sensitivity:** Fine-tuning was needed to detect hand movement accurately.
- **Power Management:** Ensuring consistent performance of all components from a single source.
- **Notification Timing:** Syncing reminders with real-world response time was crucial.

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2.5 Advantages of the Project

- **Improved Medication Adherence:** Timely reminders and monitoring reduce missed doses.
- **Real-Time Alerts:** Caregivers are notified instantly when a dose is missed.
- **Ease of Use:** Designed for non-technical users, especially elderly patients.
- **Scalable System:** Additional sensors or reminders can be integrated.
- **Educational Impact:** Offers hands-on experience in IoT, healthcare tech, and automation**.**

2.6 Shortcomings and Outcomes

While the system performs well, a few limitations remain:

- **Internet Dependency:** Blynk notifications require an active internet connection.
- **No Offline Logging:** Missed dose data is not stored during disconnections.
- **Limited Power Backup:** The system does not operate during power outages.
- **No Voice Commands Yet:** Currently under development for hands-free operation.

CHAPTER 3

HARDWARE & SOFTWARE

3.1 Introduction

This chapter outlines the hardware and software components used to develop the IoT-based Smart Home Automation System. The system uses a microcontroller (NodeMCU), relay modules, and a mobile app via Blynk IoT platform to control up to six electrical appliances remotely.

3.2 Hardware Components

1.ESP32 Microcontroller

Description:

ESP32 is a powerful microcontroller with integrated Wi-Fi and Bluetooth, making it ideal for IoT applications.

Key Features:

- Dual-core processor
- Built-in Wi-Fi & Bluetooth
- Multiple GPIO pins

  Working:

  The ESP32 receives commands from the Blynk app and controls the display and reads input from the IR sensor. It also sends data to the mobile app for real-time feedback.

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2. IR Sensor Module

Description:

This sensor detects object proximity, particularly hand movements near the dispenser or pill area.

Working:

The IR sensor sends digital signals to ESP32 when an object is detected, triggering status updates or interaction logging.

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3. Buzzer (for Alerts):

- A buzzer or sound module is used to provide an audible alert when it's time to take the medication. It can beep for a set time period to notify the user, ensuring that the patient is reminded of the dose.

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4. Blynk App

Description:
Mobile application for setting reminders, receiving alerts, and monitoring adherence.
Usage in Project:

- Displays reminders.
- Sends notifications when medication is due or missed.
- Receives status updates from ESP32.

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5. Breadboard and Jumper Wires

Used for prototyping and connecting components in a flexible and modular way.

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6. Push Button (for User Interaction):

- A simple push button used to confirm that the user has taken their medication. Pressing the button during the reminder period sends a confirmation notification to the system.

3.3 Software Tools and Libraries

1. Arduino IDE

Used to write and upload firmware to the ESP32 board.

2. Blynk IoT Platform

Provides a mobile interface with virtual pins for interaction and notifications.

3. Required Libraries:

- #include <WiFi.h>
- #include <WiFiUdp.h>
- #include <NTPClient.h>
- #include <BlynkSimpleEsp32.h>
- #include <DFRobotDFPlayerMini.h>
- #include <HardwareSerial.h>

3.4 Circuit Diagram

📌

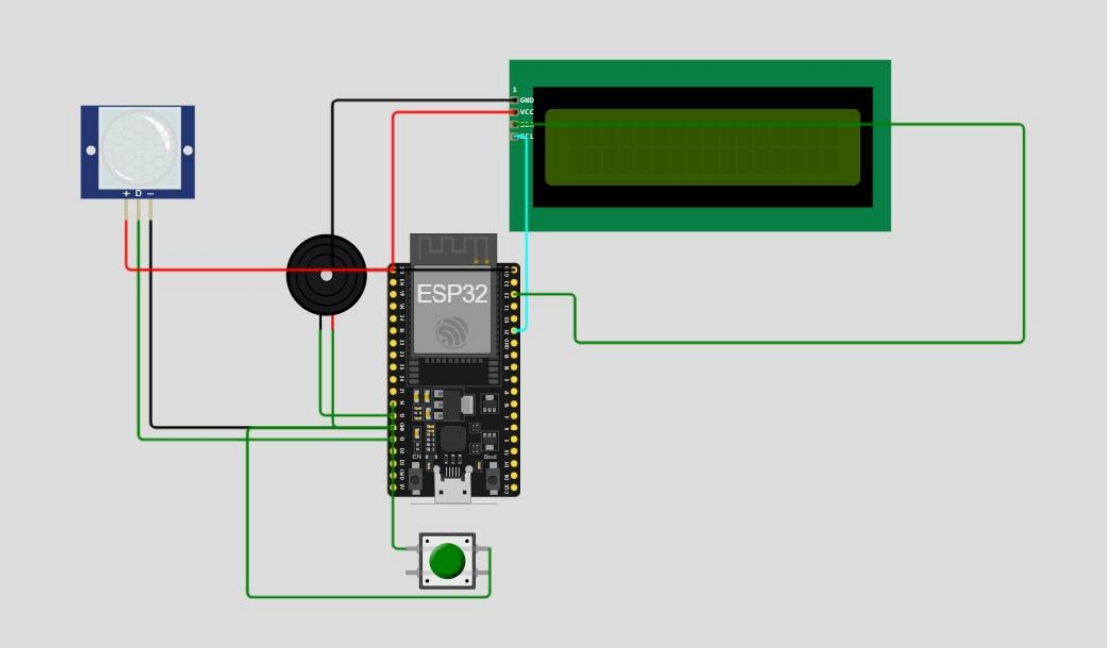


Figure 3.1 Circuit diagram

3.5 Flow Chart

📌 ***Typical Flow:***

1. System powers on

2. Connects to Wi-Fi and Blynk
3. Beeps when time for medicine
4. Waits for hand detection via IR sensor and button to be pressed
5. Sends notification and logs data via Blynk
6. Resets after action completion

3.7 Summary of Pin Connections

ESP32 Final Pin Connections

Component	ESP32 Pin	Description
Buzzer	GPIO 12	Beeps for medicine reminder
IR Sensor OUT	GPIO 13	Detects if hand is near the box
Push Button	GPIO 27	Press to confirm medicine taken
GND	GND	Common ground for all components
VCC (IR + Button + Buzzer)	3.3V or 5V	Power the components (based on their specs)

DFPlayer Mini to ESP32 Connections:

DFPlayer Mini Pin	Connects To (ESP32)	Description
VCC	5V (external or VIN)	Power supply (must be 5V)
GND	GND	Common ground
TX	GPIO 16	Connect to ESP32 RX (SoftwareSerial)
RX	GPIO 17	Connect to ESP32 TX (SoftwareSerial)
SPK1/SPK2	Speaker terminals	Connect directly to 3W speaker

CHAPTER 4

RESULTS

Overview of Results Obtained

After successfully developing and testing the Smart Medicine Box using the ESP32 microcontroller, DFPlayer Mini audio module, and Blynk IoT platform, we were able to achieve real-time medicine reminders through both **buzzer alerts** and **audio voice prompts**. The system allowed users to set up to three reminder times via the Blynk app, and based on these times, it would trigger both a buzzer sound and a voice instruction like "Time to take medicine" using the speaker.

If the user presses the button (confirming medicine intake) or interrupts the IR sensor beam (hand detection), it logs the intake on Blynk and stops the reminder. If no action is taken within 30 seconds, a **“Medicine Missed”** event is logged automatically.

Key Results

- Real-time **IoT-based medicine reminders** with **dual feedback (buzzer + speaker)**.
- Accurate **time synchronization using NTPClient** for reliable medicine scheduling.
- Interactive **Blynk interface** for setting times and viewing logs (taken/missed).
- **Log events (via Blynk)** help caretakers track medicine adherence.
- Power supplied via **5V adapter** to support ESP32 and DFPlayer simultaneously.

Screenshots of Blynk IoT Platform

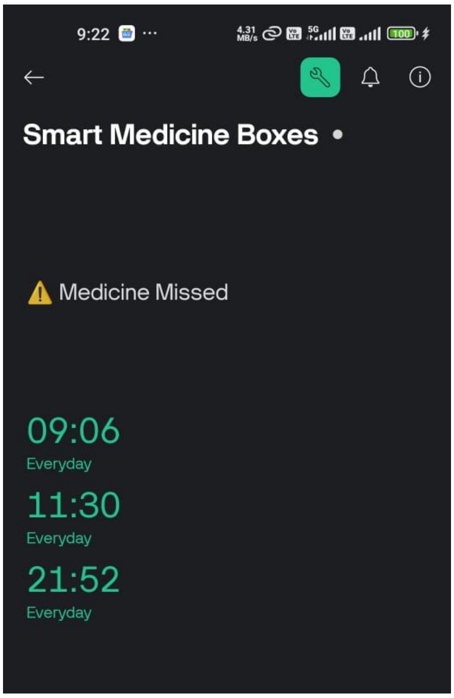


Figure 4.1:Blynk Dashboard showing V3, V4, V5 Time Input widgets.

Pictures of Working Project

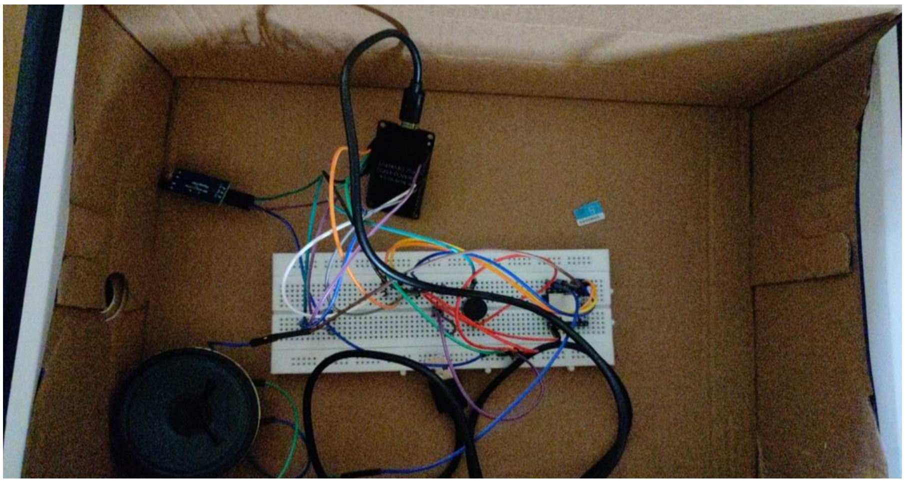
- 

Figure 2 Project (top view)

Application Areas

This project has wide applications, especially in healthcare and elderly care. Some key application domains include:

1. **Home Healthcare** – Elderly people can use it to maintain medicine schedules.
2. **Hospitals & Clinics** – To manage doses for patients in wards.
3. **Pharmacy-integrated Kits** – Smart kits provided with prescribed medication.
4. **Chronic Disease Management** – Ensuring regular medication for diabetes, BP, etc.
5. **Nursing Homes & Rehabilitation Centers** – To reduce human dependency.

Comparison with Related Work

Criteria	**Existing Systems**	**Smart Medicine Box (Our Project)**
Reminder Mode	Only sound/Blinking LEDs	Sound + Voice + IoT Notification
Cloud Sync	Not available or premium	Free with Blynk
Real-time Clock	RTC Module (extra hardware)	NTPClient (internet-synced)
Custom Schedule Setup	Fixed or manual	App-based time input (dynamic)
Logging and Monitoring	Absent	Logged on Blynk
Multi-Time Alerts	Limited	Supports 3 reminders per day

Advantages

- **Affordable and easy to build** using ESP32 and free platforms.

- **Dual Alert System** – Ensures higher probability of user response.
- **Scalable Design** – More reminder times or medicine boxes can be added.
- **Blynk Integration** – Enables remote monitoring and logging.
- **Portable and Rechargeable** – Can run using a phone charger.

Disadvantages

- Requires **Wi-Fi connection** to sync time and send notifications.
- **DFPlayer module** may be sensitive to voltage and serial connection issues.
- SD card setup for speaker audio files needs **FAT32 format** and proper folder structure.
- **No feedback if medicine is thrown** away (only button or IR detected).

Justification of Results

Compared to related IoT-based health reminder systems, our project performs better due to:

- Real-time alerts with **redundancy** (sound + text + app).
- Custom time setting using **Blynk TimeInput** widgets, unlike static alarms.
- Real-world feedback via **physical sensors and logs**, which can be accessed from anywhere.

Our testing involved setting multiple alarms, triggering responses with both hand gestures and buttons, and validating that the buzzer, speaker, and Blynk platform all functioned correctly in sync.

CHAPTER 5

Conclusion and Future Scope

Conclusion

The **Smart Medicine Box using IoT** was successfully designed and implemented to assist individuals—especially the elderly and patients with chronic illnesses—in managing their medication schedules. This project leverages the capabilities of the **ESP32 microcontroller**, **DFPlayer Mini module**, **IR sensor**, **buzzer**, and **Blynk IoT platform** to create a practical, real-time medicine reminder system. The device generates audio alerts through a speaker and visual/audio indicators via a buzzer to notify users at predetermined times. Moreover, integration with a smartphone through Blynk allows remote notifications, enabling caretakers or family members to monitor medicine intake from anywhere.

During testing, the system consistently responded to time-based triggers and sensor inputs, playing an audio message stored on an SD card and sending confirmation via the IoT platform. It was able to differentiate between cases where the medicine was taken (detected via IR sensor and weight sensor combination) and when it was missed, logging both events accurately. The device's modular and flexible architecture makes it easy to scale and customize based on the user's specific needs.

In comparison to related projects, this smart box stands out due to its **dual-feedback system** (auditory and mobile-based alerts), its **real-time monitoring**, and its **simplicity in design**. Many similar projects rely solely on alarms or app-based notifications, which can be missed or ignored. Our project enhances compliance by engaging multiple senses and by notifying both the user and caregivers.

Advantages:

- Real-time and remote monitoring of medicine intake.
- Effective for the elderly and individuals with memory disorders.
- Cost-effective, using widely available components.
- Can reduce hospitalizations due to missed doses.

Disadvantages:

- Limited to one medicine compartment in its current form.
- Dependent on internet connectivity for Blynk notifications.
- Requires initial setup and configuration that may not be user-friendly for some.

Nevertheless, this project lays a strong foundation for the integration of **IoT in healthcare**, promoting independence among patients and offering relief to caregivers.

Future Scope

The Smart Medicine Box, while effective in its current version, holds substantial potential for further development and commercial deployment. With growing reliance on smart health technologies, the device can evolve in the following directions:

1. **Multi-Compartment Scheduling:**
    - Introduce a rotating tray or sliding compartment design controlled by servo motors to manage multiple medicines with different timings.
    - Each compartment could be programmed with individual schedules, allowing for complex medication routines.
2. **Custom Mobile Application:**
    - Develop a dedicated mobile app (iOS/Android) with a more intuitive user interface than Blynk.
    - Features such as medicine history logs, user profiles, emergency contacts, and AI-generated health tips could be incorporated.
3. **GSM and GPS Integration:**
    - Adding a GSM module would allow the box to send SMS alerts or make calls to caretakers if a medicine dose is missed.
    - GPS could help in tracking mobile units for field patients or elderly individuals living alone.
4. **Voice Assist and Recognition:**

- Using voice assistants (like Google Assistant integration) or offline voice recognition, users could ask for next doses or get spoken alerts in local languages.

5. **Advanced Sensors and Analytics:**
    - Employing temperature and humidity sensors could ensure that the medicine is stored in optimal conditions.
    - Collect and visualize long-term data to identify patterns in adherence or reasons for missed doses.
6. **Camera-based Monitoring:**
    - A small integrated camera could verify medicine intake visually.
    - Caretakers can receive snapshots or live feed through cloud services for higher reliability.
7. **Battery Backup and Solar Power:**
    - To ensure continued operation during power outages, the system can be backed up by rechargeable batteries or even solar panels for rural settings.
8. **RFID or Biometric Authentication:**
    - For multi-user environments, adding fingerprint scanners or RFID tags can personalize schedules and prevent accidental intake of others' medicine.
9. **Integration with Hospitals and EHR Systems:**
    - The device could be linked with electronic health record (EHR) systems used by hospitals, automatically updating records with medicine compliance data.
10. **Commercialization and Healthcare Tie-Ups:**

- Collaborate with pharmacies or healthcare providers to deliver pre-loaded smart boxes based on prescriptions.
- Provide subscription-based services for remote monitoring and analytics.

Learnings and Technologies Used

Learnings:

Throughout the development of the **Smart Medicine Box using IoT**, we gained valuable technical and practical knowledge that extended beyond textbooks and classrooms. Key learnings included:

- **Hands-on IoT Experience:** Working with sensors, modules, and real-time data exchange through the Blynk platform helped us understand the practical implementation of IoT.
- **Embedded Systems Programming:** Writing code for the ESP32 microcontroller taught us about GPIO pin management, time-based task scheduling, and interrupt handling.
- **Serial Communication:** We learned how serial communication works between microcontrollers and external modules like DFPlayer Mini and sensors.
- **Hardware Debugging:** We encountered and resolved real-world issues like power supply mismatches, SD card format problems, and module miscommunication.
- **System Integration:** The ability to combine hardware, software, and cloud services into a single functioning system was a key takeaway.
- **User-centric Design:** We learned the importance of designing for usability and accessibility, especially considering elderly or medically vulnerable users.

Technologies Used:

- **Microcontroller:** ESP32 for its WiFi capabilities and multiple GPIO pins.
- **Programming Language:** C/C++ using Arduino IDE.
- **Sensors:** IR sensor for presence detection, weight sensor for medicine tracking.
- **Modules:**
    - DFPlayer Mini for audio playback
    - Buzzer for audio alerts

- **IoT Platform:** Blynk (mobile-based IoT dashboard)
- **Power Supply:** External 5V via adapter, with potential for battery backup.
- **Storage:** SD card (FAT32 formatted) for storing MP3 reminder messages.
- **Cloud Connectivity:** WiFi-based notifications using Blynk server.

Shortcomings and Potential Improvements

While the project achieved its primary goals, certain limitations and areas of improvement were identified:

Shortcomings:

- **Single Compartment Limitation:** The current prototype supports only one medicine type or dose at a time.
- **Manual Time Configuration:** The system lacks an RTC (Real-Time Clock) module; it relies on WiFi-based time or hardcoded timers.
- **No Backup Power:** The device fails during power outages without battery support.
- **Voice Playback Reliability:** DFPlayer Mini is sensitive to SD card formatting and wiring, making the audio function less stable.
- **Dependency on WiFi:** The entire notification system collapses if internet connectivity is lost.

Potential Improvements:

- **Multi-Medicine Support:** Include a rotating or sliding dispenser for handling multiple doses.
- **Battery and Solar Backup:** Add rechargeable batteries or solar cells for 24/7 reliability.
- **RTC Integration:** Use a real-time clock module for more accurate time tracking even during restarts.
- **Custom Mobile App:** Replace Blynk with a more user-friendly, custom-built mobile application.

- **AI-Powered Health Insights:** Integrate analytics to give users or caregivers suggestions and logs.
- **Voice Commands & Responses:** Introduce offline voice assistants for added interactivity.

Benefits for Students and Society

For Students:

- **Practical Learning:** This project served as a hands-on opportunity to apply classroom concepts in electronics, IoT, and embedded systems.
- **Problem-Solving Skills:** Students learned how to debug circuits, resolve library conflicts, and manage real-time hardware issues.
- **Teamwork and Collaboration:** The project promoted coordination among team members with diverse technical roles.
- **Portfolio Strength:** A completed smart healthcare system significantly boosts a student's resume and career readiness in IoT or embedded domains.
- **Research and Innovation:** Encouraged critical thinking and creativity for real-world applications.

For Society:

- **Assistance to Elderly and Patients:** Helps those who struggle with memory or managing multiple medications.
- **Healthcare Automation:** Reduces human error and dependence on manual caregivers.
- **Remote Monitoring:** Enables family members or healthcare providers to monitor medicine adherence remotely.
- **Cost-Effective Solution:** Uses readily available components to offer a low-cost healthcare device.
- **Potential for Rural Impact:** With minimal customization, the device can be deployed in remote areas with limited access to healthcare professionals.

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APPENDIX A

Source Code of the Project :

```
#define BLYNK_TEMPLATE_ID "TMPL3ub6W8fyh"
#define BLYNK_TEMPLATE_NAME "Smart Medicine Boxes"
#define BLYNK_AUTH_TOKEN "Hs0Ei32wxm2Zk-eXaJJ2VR9NmrZimLsE"

#include <WiFi.h>
#include <WiFiUdp.h>
#include <NTPClient.h>
#include <BlynkSimpleEsp32.h>
#include <DFRobotDFPlayerMini.h>
#include <HardwareSerial.h>

char ssid[] = "shivek";
char pass[] = "ZOADIACKILLER";

// Pins
#define BUZZER_PIN 12
#define IR_PIN 13
#define BUTTON_PIN 27

// DFPlayer connected to RX2 (GPIO16), TX2 (GPIO17)
HardwareSerial mySerial(2);
DFRobotDFPlayerMini dfplayer;

int medicineHour1 = -1, medicineMinute1 = -1;
int medicineHour2 = -1, medicineMinute2 = -1;
int medicineHour3 = -1, medicineMinute3 = -1;

BlynkTimer timer;
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP, "pool.ntp.org", 19800, 60000); // IST

bool medicineDue = false;
```

```
bool medicineTaken = false;

void setup() {
  Serial.begin(115200);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(IR_PIN, INPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timeClient.begin();

  mySerial.begin(9600, SERIAL_8N1, 16, 17); // RX=16, TX=17
  if (!dfplayer.begin(mySerial)) {
    Serial.println("DFPlayer not connected!");
  } else {
    dfplayer.volume(25); // Volume: 0–30
  }

  timer.setInterval(30000L, checkMedicineTime); // every 30 seconds
}

void loop() {
  Blynk.run();
  timer.run();
}

// Time Inputs
BLYNK_WRITE(V3) {
  TimeInputParam t(param);
  medicineHour1 = t.getStartHour();
  medicineMinute1 = t.getStartMinute();
}
BLYNK_WRITE(V4) {
  TimeInputParam t(param);
  medicineHour2 = t.getStartHour();
  medicineMinute2 = t.getStartMinute();
```

```
}
BLYNK_WRITE(V5) {
  TimeInputParam t(param);
  medicineHour3 = t.getStartHour();
  medicineMinute3 = t.getStartMinute();
}

void checkMedicineTime() {
  timeClient.update();
  int currentHour = timeClient.getHours();
  int currentMinute = timeClient.getMinutes();

  if ((currentHour == medicineHour1 && currentMinute == medicineMinute1) ||
      (currentHour == medicineHour2 && currentMinute == medicineMinute2) ||
      (currentHour == medicineHour3 && currentMinute == medicineMinute3)) {
    if (!medicineDue) {
      startMedicineAlert();
    }
  }
}

void startMedicineAlert() {
  medicineDue = true;
  medicineTaken = false;

  Blynk.logEvent("medicine_time", "⏰ Time to take your medicine!");

  // Play voice message (e.g., 0001.mp3)
  dfplayer.play(1);

  // Beep for 30 seconds
  unsigned long startTime = millis();
  while (millis() - startTime < 30000) {
    digitalWrite(BUZZER_PIN, HIGH);
    delay(150);
    digitalWrite(BUZZER_PIN, LOW);
```

```
    delay(150);

    if (digitalRead(BUTTON_PIN) == LOW) {
      medicineTaken = true;
      break;
    }
  }

  if (medicineTaken) {
    Blynk.virtualWrite(V0, "✅ Medicine Taken");
    Blynk.virtualWrite(V2, 1);
    Blynk.logEvent("medicine_taken", "✅ Medicine Taken");
    Serial.println("Medicine Taken");
  } else {
    Blynk.virtualWrite(V0, "⚠️ Medicine Missed");
    Blynk.logEvent("medicine_missed", "⚠️ Medicine Missed");
    Serial.println("Medicine Missed");
  }

  medicineDue = false;
  delay(3000);
}
```